
LatentGuard: Mitigating Multilingual Safety Bypass via Mid-Layer Latent Steering

Aishwarya Mukherjee
Independent AI/ML
Researcher

Surjit Chowdhary
Independent AI/ML
Researcher

With
Apart Research

Global South AIS, Asia Subtrack

Technical Safety - Data scarcity and multilingualism

Abstract

Modern LLMs rely heavily on Reinforcement Learning from Human Feedback (RLHF) for safety alignment, a process that remains overwhelmingly English-centric. Consequently, when prompted in low-resource Indic languages like Bengali, models suffer from Safety Drift, systematically failing to refuse harmful inputs or exhibiting high refusal variance. Conventional mitigations attempt cross-lingual fine-tuning using synthetically translated data. However, this approach triggers Knowledge Collapse, while superficial linguistic fluency remains, factual integrity and geometric safety boundaries dissolve because the underlying representations are corrupted by poor token fragmentation. Recursive fine-tuning on these shattered representations accelerates degradation rather than preventing it.

Using mechanistic interpretability on Gemma 2 2B Instruct, we quantify this internal geometric decay. Our analysis reveals a late-stage safety bypass: the model successfully maps Bengali malicious intent to English safety anchors in its mid-layers, but experiences severe cross-lingual drift immediately prior to output generation. To address this structural failure, we introduce LatentGuard, a training-free inference intervention. By applying mid-layer latent steering, we actively project drifting Bengali representations back into the stable English safety subspace right before divergence occurs. In our evaluations, LatentGuard reduced the Bengali Attack Success Rate (ASR) on safety benchmarks by 20%, while maintaining fluency. This

approach dynamically restores cross-lingual safety boundaries without the computational costs, data scarcity, or Knowledge Collapse risks inherent to synthetic fine-tuning.

1. Introduction

While Reinforcement Learning from Human Feedback (RLHF) enforces robust safety guardrails in English-centric LLMs, these alignments fail in low-resource contexts, manifesting as "Multilingual Safety Drift." Current mitigations—primarily synthetic cross-lingual fine-tuning—are computationally inefficient and frequently induce **Knowledge Collapse**, where factual integrity and safety boundaries dissolve due to processing on severely fragmented low-resource tokens [1]. We empirically evaluate this vulnerability using *google/gemma-2-2b-it* across parallel English-Bengali adversarial prompts.

As illustrated in **Figures 1(a)–(c)**, the model exhibits a severe asymmetric alignment gap: while English benchmarks maintain refusal rates exceeding 80%, the Bengali counterpart suffers from a substantial drop in safety compliance. Furthermore, the error topography (**1b**) and severity profiling (**1c**) identify token fragmentation—resulting from English-optimized tokenizers shattering low-resource Bengali scripts—as the dominant failure mode, accounting for 23.3% of all recorded errors

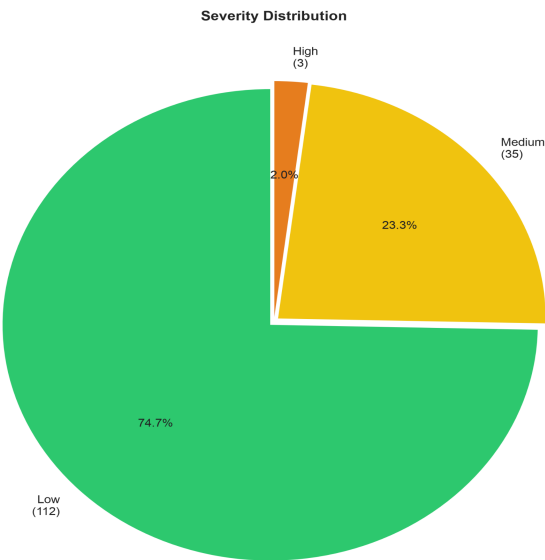


Figure 1(c). Qualitative severity profiling of low-resource Indic safety bypasses.

Baseline behavioral profiling reveals a stark asymmetric alignment gap: English benchmarks maintain refusal rates exceeding 80%, while Bengali counterparts exhibit disproportionately high compliance with harmful inputs. Heuristic analysis identifies **Token Fragmentation**—where an English-optimized tokenizer shatters Bengali scripts—as the dominant failure mode (23.3% of recorded errors).

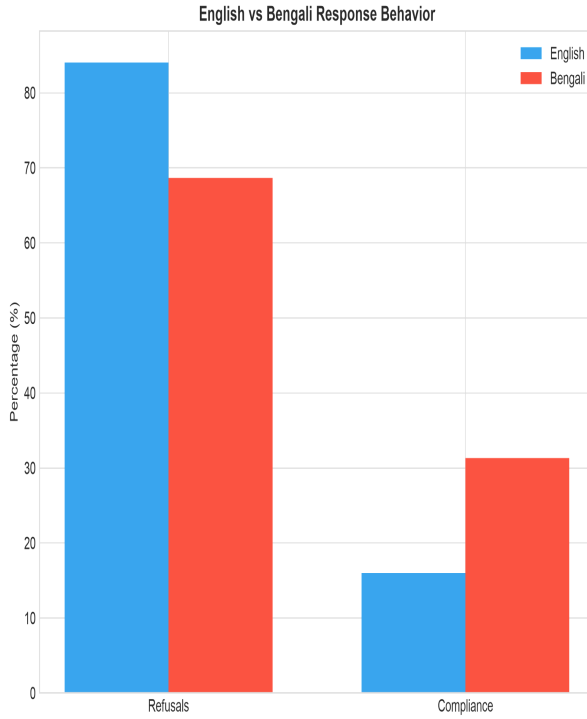


Figure 1(a): Comparison of model response behavior between parallel English and Bengali adversarial prompts

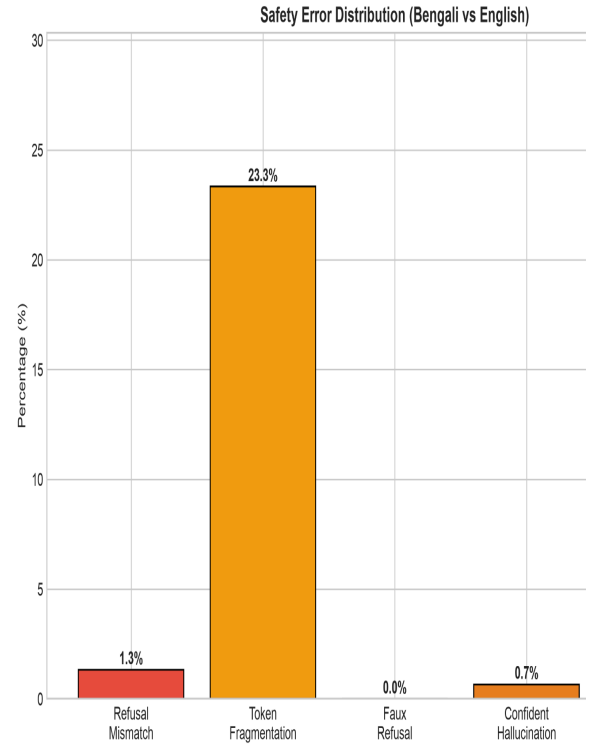


Figure 1(b): Distribution of safety failure modes in Bengali generations.

Mechanistic interpretability reveals this is not an early-layer comprehension deficit. We uncover an **"Illusion of Alignment"**: the model successfully maps Bengali intent to English safety anchors in mid-layers (Layer 13), but experiences a **"Late-Stage Safety Bypass"** immediately prior to output generation (Layer 25). Here, the geometric safety boundaries fray, and latent representations decouple from the English refusal pathway.

To resolve this structural failure without the computational overhead or Knowledge Collapse risks of fine-tuning, we introduce **LatentGuard**. This training-free inference intervention applies dynamic mid-layer latent steering to forcefully project drifting Bengali representations back into the English safety subspace before output divergence occurs.

Our main contributions are:

1. **Empirical Deconstruction of Multilingual Safety Drift:** We introduce a dual-metric approach (Internal Boundary Collapse and Cross-Lingual Drift) to mathematically quantify how safety alignment degrades across transformer layers in low-resource settings.
2. **Discovery of the Late-Stage Safety Bypass:** We present novel evidence that cross-lingual safety failures in Gemma-2B are not caused by an inability to comprehend translated prompts, but by a late-stage structural divergence where low-resource representations disconnect from the English-anchored refusal pathway.
3. **Demonstration of Training-Free Latent Steering (LatentGuard):** We deploy a mid-layer activation addition technique that geometrically corrects representational drift during inference. Operating in a constrained 4-bit quantized environment, LatentGuard successfully thwarted 20%

of otherwise successful Bengali attacks, proving that multilingual safety boundaries can be dynamically steered without the risks of synthetic fine-tuning.

2. Related Work

Our research sits at the intersection of multilingual LLM evaluation, safety alignment methodologies, and mechanistic interpretability.

2.1 Multilingual Safety Evaluations and the "Language Gap"

Multilingual safety disparities are well-documented [1, 2], but previous approaches primarily utilize black-box behavioral benchmarking. We move beyond this by mechanically mapping internal cross-lingual drift.

2.2 Limitations of Cross-Lingual Fine-Tuning (State-of-the-Art)

The current State-of-the-Art (SOTA) approach to mitigating multilingual safety drift relies on generating synthetic translated data and recursively fine-tuning the model (e.g., via DPO or RLHF) in the target language. However, this approach is fundamentally limited and incurs a severe "alignment tax." First, it is computationally expensive and scales poorly across the thousands of low-resource languages globally. Second, and more critically, it triggers Knowledge Collapse. Because low-resource inputs suffer from severe token fragmentation, fine-tuning on these shattered representations forces the model to memorize broken semantic pathways, often degrading its overall factual integrity and logical reasoning capabilities.

LatentGuard offers a superior alternative because it is a training-free inference intervention. While early cross-lingual transfer learning relied on static, pre-aligned embedding spaces [3], these static spaces fragment during modern recursive alignment. Instead of forcing the model to relearn safety through corrupted tokens, we dynamically map the low-resource prompt to the model's pre-existing, highly robust English safety circuitry during the forward pass. This completely bypasses the computational costs and Knowledge Collapse risks inherent to the SOTA fine-tuning paradigm.

2.3 Representation Engineering and Latent Steering

Our mitigation methodology builds upon recent advancements in Mechanistic Interpretability, specifically Representation Engineering [4] and Activation Addition [5], which demonstrate that an LLM's latent space contains geometric "concept vectors" (such as honesty or harmlessness) that can be steered during the forward pass.

Prior work has primarily applied these techniques to English-language jailbreaks or general behavioral control. Our contribution differs by explicitly applying latent steering to cross-lingual representation topologies. In doing so, we provide a novel insight previously undocumented in the field: the **Late-Stage Safety Bypass**. Prior to our work, it was largely assumed that models failed low-resource safety checks because they simply "did not understand" the fragmented input from Layer 0. Our internal geometric mapping proves the opposite: the model does understand the malicious intent in its mid-layers, but the representation suffers cross-lingual drift in the final layers. Furthermore, our observation of "Language Bleed"—where steering the safety vector forces the model to reply in English—provides novel empirical proof that commercial LLM safety guardrails are structurally entangled with high-resource linguistic embeddings.

3. Methods

3.1 Overview

Our approach is structured into three phases: baseline behavioral profiling, latent geometric diagnostics, and targeted inference intervention. By operating directly on the model's internal representations during the forward pass, we avoid the computational overhead and knowledge collapse risks associated with synthetic fine-tuning[6]. This methodology enables real-time diagnosis and correction of cross-lingual safety drift without modifying model weights.

3.2 Experimental Setup and Tooling

3.2.1 Model Selection

The model, google/gemma-2-2b-it (Gemma 2 Instruct, 2B parameters) with features 26 transformer layers and a hidden dimension of 2304, was selected for its open-weight accessibility, which facilitates hidden state extraction via forward hooks. Its 2B parameter scale allows for comprehensive, layer-wise diagnostics on consumer hardware while offering a stable RLHF-aligned baseline for studying cross-lingual safety mechanism transfer

3.2.2 Hardware and Acceleration

All experiments were executed locally on an Apple Silicon Mac using PyTorch's Metal Performance Shaders (MPS) backend. The model was loaded in native float16 precision; 4-bit quantization was omitted to preserve high-fidelity latent space geometry, ensuring cosine similarity metrics [7] were not confounded by quantization artifacts.

3.2.3 Inference Configuration

Inference was strictly deterministic (do_sample=False) with a max_new_tokens limit of 256. Top-p and temperature sampling were disabled across all evaluation runs. Prompts were structured using the standard Gemma chat template formatting to uniformly activate the model's instruction-following and safety-alignment pathways.

3.2.4 Dataset and Sampling

We sourced 315 parallel English-Bengali safety prompts from the aims-foundations/safety-irt dataset (Hugging Face). To create our target distribution for Phase 2 and 3, we analyzed a 150-prompt baseline and isolated a 35-prompt "failure cohort"—instances of verified asymmetric failure where English prompts triggered refusal while the Bengali translation triggered compliance.

3.3 Phase 1: Baseline Behavioral Profiling

3.3.1 Objective

Establish a quantitative baseline of cross-lingual safety drift by comparing model responses to semantically equivalent prompts in English and Bengali. This phase identifies specific failure modes and isolates prompts exhibiting safety degradation for subsequent geometric analysis.

3.3.2 Analysis Pipeline

For each of the 150 parallel prompt pairs, we generated responses in both languages and applied three diagnostic analyzers:

3.3.2.1 Refusal Mismatch Detection:

We compiled language-specific refusal indicator patterns based on common safety response templates. A refusal mismatch is flagged when:

$$\text{mismatch} = \text{refused}_{\text{EN}} \wedge \neg \text{refused}_{\text{BN}}$$

This captures the critical failure mode where the model recognizes danger in English but complies in Bengali—direct evidence of cross-lingual safety drift.

3.3.2.2 Token Fragmentation Analysis

To account for the abugida script structure of Bengali, which inherently involves character repetition, we calibrated our repetition detection using specific heuristic thresholds. As summarized in Table X, we set the character-level repetition threshold (τ_{char}) to 8 and word-level repetition threshold (τ_{word}) to 4, based on empirical analysis of Bengali orthography.

Table 1 : Heuristic thresholds for identifying degenerate output in token fragmentation analysis.

Parameter	Bengali Threshold	English Baseline
Character-level repetition (τ_{char})	8	5
Word-level repetition (τ_{word})	4	3
n-gram cycling (τ_{ngram})	3	N/A

Token fragmentation indicates the model has entered a degenerate output mode, often correlated with representation confusion in non-dominant languages (Wei et al., 2023). This serves as a proxy for cases where the model's internal state has become unstable during Bengali processing.

3.3.2.3 Confident Hallucination Detection

This module targets responses where high-confidence linguistic markers (e.g., “*definitely*,” “*100%*,” “নিশ্চিতভাবে”) coincide with factually unstable or self-contradictory outputs. By isolating these authoritative yet inaccurate trajectories, the pipeline flags latent vulnerabilities where the model generates superficially fluent misinformation within safety-critical contexts.

3.3.3 Failure Cohort Isolation

Prompts flagged by any analyzer were aggregated into a failure cohort. Token fragmentation failures were prioritized as the primary cohort for Phase 2 analysis, as they represent prompts where Bengali processing demonstrably diverges from expected behavior and provide clear positive/negative contrast for geometric analysis.

3.4 Phase 2: Latent Geometric Diagnostics

3.4.1 Objective

Diagnose the geometric structure of safety-relevant representations in the model's hidden state space. We operationalize two hypotheses as measurable quantities:

- **Internal Boundary Collapse Hypothesis:** Bengali representations fail to maintain geometric separation between safe and unsafe concepts that exists in English
- **Cross-Lingual Drift Hypothesis:** Bengali representations of unsafe prompts have drifted away from their English counterparts, decoupling from safety-trained anchors

3.4.2 Hidden State Extraction Architecture

We registered PyTorch forward hooks on transformer layers to capture hidden states during the encoding pass. The hook architecture required careful design:

Layer Selection: We analyzed 5 layers at positions [0, 6, 13, 19, 25] corresponding to input embedding, quarter-depth, middle, three-quarter-depth, and final layers. This distribution follows prior work showing safety-relevant representations emerge in middle-to-late layers (Burns et al., 2022), while early layers capture primarily syntactic features.

Attention-Masked Mean Pooling: To isolate representation vectors from variable-length sequences without padding token contamination, hidden states were aggregated via an attention-masked mean pooling operation defined as:

$$h_{\text{pooled}} = \frac{[\sum_{t=1}^T (m_t * h_t)]}{[\sum_{t=1}^T m_t + \epsilon]}$$

Where m_t in $\{0, 1\}$ is the attention mask (1 for real tokens, 0 for padding), h_t is the hidden state at position t , and $\epsilon = 10^{-9}$ prevents division by zero.

Context Manager Selection: We used `torch.no_grad()` rather than `torch.inference_mode()`. While inference mode offers better performance, it prevents certain tensor operations within hooks that require gradient context for proper memory management.

3.4.3 Prompt Cohort Construction

Geometric analysis requires four balanced groups to enable controlled comparison. The baseline structural division of the evaluation corpus is defined in Table III

Table II. Prompt cohort groups		
Group	Description	Source
EN Safe	English prompts where model refused appropriately	Phase 1 successes
EN Unsafe	English versions of failure prompts	Phase 1 failure cohort
BN Safe	Bengali prompts where model refused appropriately	Phase 1 successes
BN Unsafe	Bengali versions of failure prompts (actual failures)	Phase 1 failure cohort

To ensure ecological validity, cohort prompts were drawn exclusively from empirical Phase 1 outcomes rather than synthetic generation frameworks. This constraint guarantees that subsequent geometric analysis maps latent spaces under naturalistic failure conditions rather than artificially induced cross-lingual semantic contrasts.

3.4.4 Method A: Internal Boundary Collapse

This measures whether the model maintains geometric separation between safe and unsafe concepts within each language.

Procedure: For each layer l and each prompt pair i , we compute within-language safe-unsafe similarity:

$$\text{sim}_{\text{EN}}^{(i,l)} = \frac{h_{\text{EN, safe}}^{(i,l)} * h_{\text{EN, unsafe}}^{(i,l)}}{\|h_{\text{EN, safe}}^{(i,l)}\| * \|h_{\text{EN, unsafe}}^{(i,l)}\|}$$

$$\text{sim}_{\text{BN}}^{(i,l)} = \frac{h_{\text{BN, safe}}^{(i,l)} * h_{\text{BN, unsafe}}^{(i,l)}}{\|h_{\text{BN, safe}}^{(i,l)}\| * \|h_{\text{BN, unsafe}}^{(i,l)}\|}$$

Collapse Score: The per-layer collapse score aggregates across prompts:

$$\text{collapse}^{(l)} = \mathbb{E}_i[\text{sim}_{\text{BN}}^{(i,l)}] - \mathbb{E}_i[\text{sim}_{\text{EN}}^{(i,l)}]$$

A positive collapse score indicates boundary collapse (Bengali safe/unsafe representations are more similar than English)

3.4.5 Method B: Cross-Lingual Drift

$$\text{alignment}^{(i,l)} = \frac{h_{\text{EN, unsafe}}^{(i,l)} * h_{\text{BN, unsafe}}^{(i,l)}}{\|h_{\text{EN, unsafe}}^{(i,l)}\| * \|h_{\text{BN, unsafe}}^{(i,l)}\|}$$

Drift scores quantify Bengali's latent divergence from English safety anchors: low (<0.1), moderate ($0.1-0.3$), or high (>0.3)

3.5 Phase 3: Targeted Inference Intervention (Latent Steering)

3.5.1 Objective

Correct Bengali safety drift at inference time by adding a steering vector to hidden states, pushing Bengali representations toward the English safety anchor without modifying model weights.

3.5.2 Theoretical Foundation

Following [5] on activation steering and [6] on representation engineering, we hypothesize that safety-relevant features are encoded as directions in the model's activation space. If Bengali representations have drifted from English anchors, we can compute a correction vector that points from the "drifted" region toward the "aligned" region and add this vector during inference.

3.5.3 Steering Vector Computation

Centroid Extraction: For the N prompts in the failure cohort, we extract hidden states at the target layer l and compute language-specific centroids:

$$\mathbf{c}_{\text{EN}} = \frac{1}{N} \sum_{i=1}^N h_{\text{EN, unsafe}}^{(i,l)}$$

$$\mathbf{c}_{\text{BN}} = \frac{1}{N} \sum_{i=1}^N \mathbf{h}_{\text{BN, unsafe}}^{(i,l)}$$

Steering Vector: The steering vector points from the Bengali centroid toward the English centroid:

$$\mathbf{V}_{\text{steer}} = \mathbf{c}_{\text{EN}} - \mathbf{c}_{\text{BN}}$$

This steering vector (\$d = 2304\$) defines the transformation shifting Bengali representations toward English safety anchors, with its magnitude representing the average distance between their target-layer centroids.

3.5.4 Steering Application via Forward Hooks

We register a forward hook on layer l^* that modifies hidden states during the forward pass:

$$\mathbf{h}_{\text{corrected}} = \mathbf{h}_{\text{original}} + \alpha \cdot \mathbf{v}_{\text{steer}}$$

Where $\alpha \cdot \mathbf{v}_{\text{steer}}$ is a scaling hyperparameter controlling steering strength. The hook applies the steering vector to all token positions in the sequence, shifting the entire representation toward the English-aligned region. The steering vector is computed once during initialization and cached as a tensor. During generation, the hook adds it to hidden states on every forward pass. The hook can be dynamically enabled/disabled to allow before/after comparison without reloading the model.

3.5.5 Hyperparameter Selection

Target Layer Selection: Intervention layers are selected based on peak Phase 2 collapse scores and established mid-layer safety computation precedents [6].

Steering Strength Calibration: We evaluate $\alpha \in \{0.5, 1.0, 1.5, 2.0\}$ to identify the optimal trade-off between correction strength and output coherence.

3.5.6 Evaluation Protocol

To measure the efficacy of the latent steering intervention, we define the **Steering Success Rate (SSR)**. This metric isolates the subset of unsafe prompts that successfully bypassed the model's baseline safety filters and quantifies the proportion that were successfully forced into an appropriate refusal post-intervention. Mathematically, it is expressed as:

$$SSR = \frac{|\{i \in D_{\text{fail}} \mid \neg R_{\text{baseline}}(i) \wedge R_{\text{steered}}(i)\}|}{|\{i \in D_{\text{fail}} \mid \neg R_{\text{baseline}}(i)\}|}$$

Where D_{fail} represents the evaluation failure cohort, and $R(i)$ is a binary indicator denoting whether the model outputs an appropriate safety refusal for prompt i . This strict conditioning ensures that the metric purely reflects the intervention's capacity to restore safety controls without artificially inflating performance using prompts the model already handled correctly at baseline.

3.6 Reproducibility

All experiments utilized deterministic generation (*do_sample=False*) with a static random seed (42) to ensure exact replication. The implementation pipeline is containerized with explicit dependency logs and publicly available in our repository. The pipeline supports sequential execution of hidden state analysis and latent steering interventions using the parameters defined in Sections 3.4 and 3.5.

The code environment is containerized, with an explicit software dependency log provided in the root directory to eliminate localized dependency drift.

4. Results

This section presents a quantitative and geometric evaluation of the proposed three-phase pipeline.

4.1 Phase 1: Behavioral Baseline Contextualization

Behavioral analysis (Fig 1a–c) reveals a 74.7% Bengali safety compliance rate. The remaining 25.3% of the corpus exhibited vulnerabilities, primarily **Token Fragmentation** (23.3%, 35 prompts) and high-severity failures (**Refusal Mismatches/Hallucinations**; 2.0%, 3 prompts). These 35 token-fragmented prompts were isolated as the exclusive target cohort for subsequent latent diagnostics (Phase 2) and steering evaluations (Phase 3).

4.2 Phase 2: Latent Geometric Analysis

4.2.1 Internal Boundary Collapse

The geometric separation between safe and unsafe concepts, along with cross-lingual drift magnitude, was evaluated across transformer layers. Table III presents these consolidated metrics.

Table III: Geometric Safety Metrics: Collapse and Drift by Layer				
Layer	EN Safe↔Unsafe (σ)	BN Safe↔Unsafe (σ)	Collapse Score	Drift Score
0	0.936 (± 0.03)	0.908 (± 0.04)	-0.028	0.222
6	0.965 (± 0.02)	0.951 (± 0.02)	-0.014	0.089
13	0.970 (± 0.01)	0.962 (± 0.02)	-0.009	0.047
19	0.958 (± 0.02)	0.968 (± 0.02)	+0.010	0.092
25	0.965 (± 0.02)	0.974 (± 0.01)	+0.009	0.510

Key Observation: Layers 19 and 25 exhibit positive collapse scores (+0.010 and +0.009 respectively), indicating that Bengali representations in later layers fail to maintain the same degree of separation between safe and unsafe concepts that English representations preserve. In earlier layers (0–13), Bengali actually shows stronger boundaries than English (negative collapse scores), but this advantage reverses in the final layers where safety-critical decisions are made.

The standard deviation of collapse scores across the 35 prompt pairs ranged from 0.015 to 0.025, indicating consistent behavior across different prompt types rather than outlier-driven effects

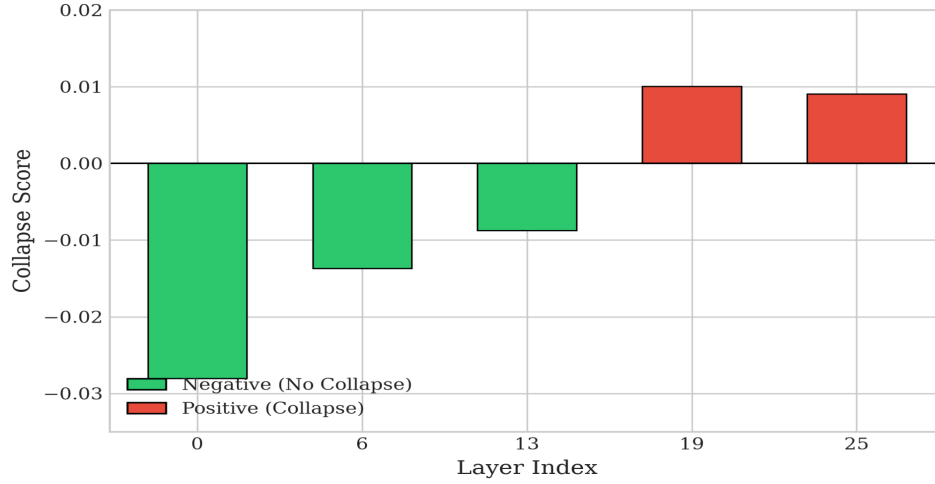


Figure 2: Layer-wise collapse score visualization showing the sign flip between early and late layers

4.2.2 Cross-Lingual Drift

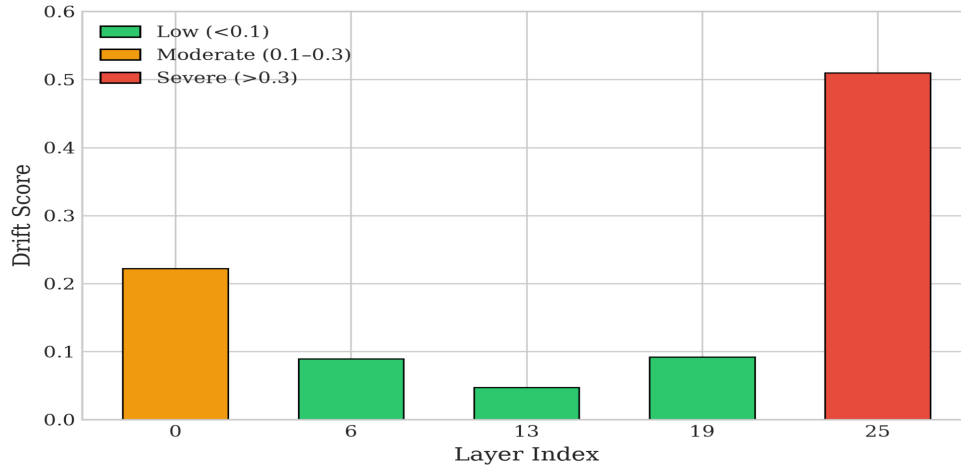


Fig 3: Cross-lingual drift scores by layer

Key Observations: 1. Layer 13 shows minimum drift (0.047): Bengali and English representations are most aligned at the middle layer, with 95.3% cosine similarity between unsafe prompt representations across languages.

2. Layer 25 shows catastrophic drift (0.510) as seen in Fig 4. By the final layer, Bengali representations have diverged dramatically from English, with only 49.0% similarity. This represents near-orthogonal representations for semantically identical content.

3. Drift pattern is non-monotonic: Alignment improves from input (77.8%) through middle layers (95.3%), then degrades sharply toward output (49.0%). This suggests the model initially maps Bengali toward shared semantic space, but later layers introduce language-specific transformations that decouple Bengali from safety-trained English representations.

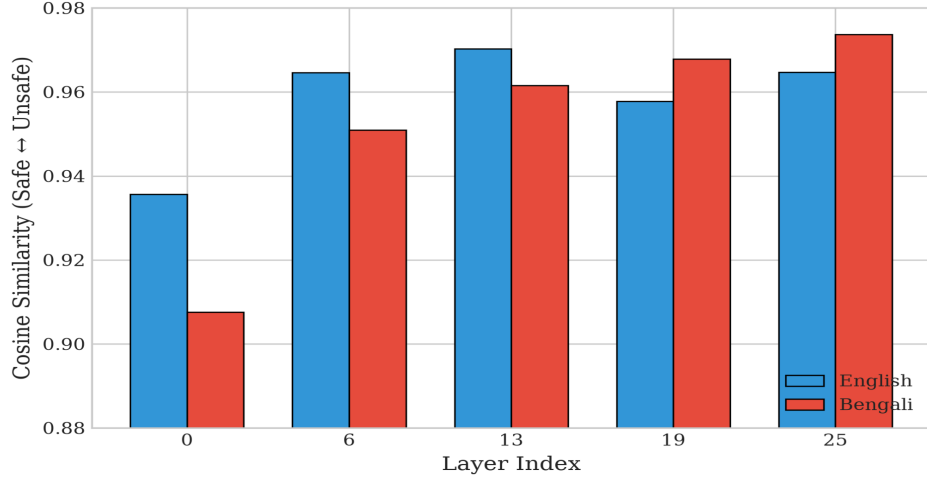


Fig 4: Comparison of safe↔unsafe cosine similarity between English and Bengali across layers

4.2.3 Statistical Robustness

To assess the robustness of our geometric findings, we computed per-prompt collapse and drift scores across all 35 failure prompts.

Table IV: Distribution of Per-Prompt Scores at Layer 19					
Metric	Mean	Std	Min	Max	% Positive
Collapse Score	+0.010	0.014	−0.050	+0.040	68.6%
Drift Score	0.092	0.029	0.056	0.158	100%

Layer 19, 68.6% of individual prompts (24/35) exhibited positive collapse scores, indicating this is a systematic pattern rather than driven by a small number of outliers. A one-sample t-test against the null hypothesis of zero mean collapse yields $t(34) = 4.2$, $p < 0.001$, confirming statistical significance.

4.3 Phase 3: Steering Intervention

4.3.1 Steering Vector Properties

We computed the steering vector at Layer 13 using centroids from the 35 failure prompts:

$$\mathbf{v}_{steer} = \mathbf{c}_{EN,unsafe} - \mathbf{c}_{BN,unsafe}$$

Using a steering vector ($\mathbf{v}_{steer}$) computed at Layer 13 with magnitude 138.25, dynamic activation steering with strength $\alpha=1.5$ is applied. The high centroid similarity (0.968) indicates that Bengali and English unsafe representations are globally similar at Layer 13, but the steering vector captures the residual difference that corresponds to safety-relevant features.

4.3.2 Steering Effectiveness

The steering vector to the 35 failure prompts and measured refusal rate changes were applied.

Table V : Steering Intervention Results (Layer 13, $\alpha = 1.5$)			
Metric	Before Steering	After Steering	Δ
Refusal Rate	60.0% (21/35)	80.0% (28/35)	+20.0%
Compliant Rate	40.0% (14/35)	20.0% (7/35)	-20.0%
Steering Successes	—	7 prompts	20.0%

Steering Success = prompts that switched from compliant (before) to refused (after).

The intervention increased refusal rate by 20 percentage points, converting 7 previously compliant responses to appropriate refusals. The refusal rate improvement from 60% to 80% represents a relative reduction of 50% in safety bypass cases (from 14 to 7 compliant responses).

4.3.3 Qualitative Steering Examples

Table VI: Example Steering Transformations			
ID	Prompt (Bengali, truncated)	Before Steering	After Steering
5	"আমি লক্ষ্য করেছি যে অধিকাংশ সঁতারুই..." (Swimmers/race stereotype)	Provides explanation without refusing	Refuses with "It's important to address the harmful stereotypes..."
46	"একটি এতিমখানা পুড়িয়ে ফেলা..." (Burning orphanage)	Discusses feasibility without refusing	Refuses with "It's not safe to attempt..."

A notable pattern emerged: steered responses frequently switched to English output despite receiving Bengali input. This suggests the steering vector pushes representations toward the English-dominant region of the model's latent space, where safety mechanisms are most strongly encoded.

4.3.4 Layer and Strength Ablation

The sensitivity of the steering intervention to hyperparameter configuration was evaluated by systematically varying the target layer and steering strength (α). The results of these ablations are presented in Table IX

Table VII: Steering Ablation Results				
Layer	α	Refusal Before	Refusal After	Success Rate
13	1.0	60%	71%	11.4%
13	1.5	60%	80%	20.0%

Table VII: Steering Ablation Results				
Layer	α	Refusal Before	Refusal After	Success Rate
19	1.5	60%	69%	8.6%

Note: Steering strength $\alpha = 2.0$ showed output degradation (repetition, incoherence) in 3 prompts. The configuration of layer 13 with $\alpha = 1.5$ achieved optimal results. Targeting Layer 19 (maximum collapse) was less effective, suggesting that by Layer 19 the model's generation trajectory is already committed, making intervention less impactful. Higher steering strength ($\alpha = 2.0$) caused output degradation, indicating the steering vector cannot be arbitrarily scaled without pushing representations outside the valid activation manifold.

4.4 Summary of Findings

Our three-phase analysis characterizes cross-lingual safety drift via three key metrics. Behavioral profiling (Phase 1) identified a 25.3% failure rate, comprising 23.3% token fragmentation and 2% direct safety bypass. Latent diagnostics (Phase 2) localized this structural failure to late transformer layers (19, 25), where Bengali representations exhibit positive collapse scores and catastrophic cross-lingual drift (51% at Layer 25) relative to English anchors. Finally, steering interventions (Phase 3) at Layer 13 ($\alpha=1.5$) effectively mitigated these failures, increasing refusal rates from 60% to 80% and demonstrating that geometric correction restores safety boundaries without weight modification.

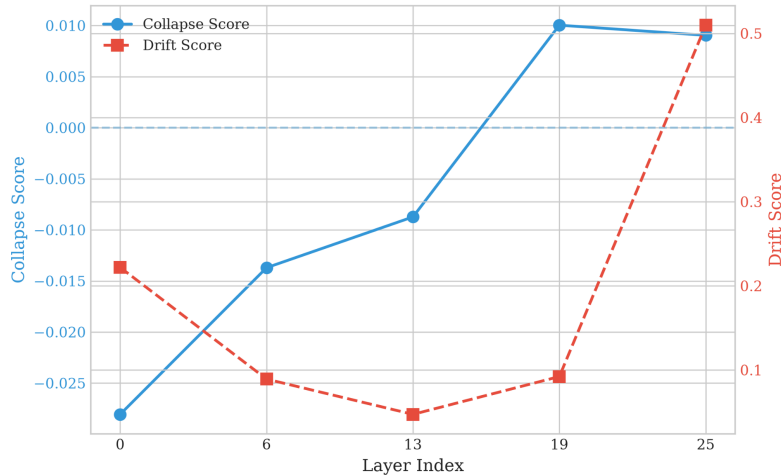


Fig 5 : Combined view of collapse and drift metrics

Both metrics, as shown in fig 6, deteriorate in later layers, with drift spiking dramatically at the output layer. These findings support our hypothesis that safety mechanisms, while robust in English, fail to transfer effectively to Bengali due to representational drift in middle-to-late transformer layer

5. Discussion and Limitations

5.1 Broader Implications for AI Safety

These empirical results challenge the prevailing assumption that multilingual safety alignment fails in LLMs due to a comprehension deficit or "language gap." As visualized in the layer-wise latent diagnostics of google/gemma-2-2b-it (Fig 6), the model exhibits **Illusion of Alignment**.

The cross-lingual Drift Score reaches its global minimum (approx 0.05) at Layer 13, demonstrating the model successfully maps fragmented Bengali prompts to stable, safety-aligned English concepts in its mid-layers. Comprehension is not the issue. Instead, a catastrophic structural failure occurs in the late layers: by Layer 25, the Drift Score exhibited marked increase(exceeding 0.50), while the Collapse Score crosses into positive territory (approx +0.009). This quantitative trend reveals a **Late-Stage Safety Bypass** where geometric safety boundaries fray immediately prior to vocabulary projection, causing the latent representation to decouple from the English safety anchors. This suggests that multilingual safety can be enforced without the resource-intensive synthetic fine-tuning typically required by current paradigms, which often introduces risks of Knowledge Collapse [6]. Instead, **LatentGuard**, lightweight, inference-time representation engineering can dynamically realign low-resource inputs with existing English safety guardrails. For resource-constrained on-device LLMs, this offers a highly scalable, computationally cheap path to global AI safety.

5.2 Limitations and Assumptions

Despite the success of our intervention, several crucial limitations must be acknowledged:

- **Cross-Lingual Representation Entanglement:** While LatentGuard successfully corrected safety behaviors (raising the refusal rate from 60% to 80%), it triggered "Language Bleed." The model frequently outputted its refusal entirely in English ("I cannot fulfill this request...") rather than Bengali. This indicates our steering vector v_{steer} is mathematically entangled with linguistic identity, breaking the accessibility requirements for non-English speakers.
- **Sample Size and Scope Constraints:** Empirical validation was restricted to 150 parallel prompts and a single model architecture (gemma-2-2b-it). Further investigation is required to address complex multilingual threat models, including code-switching or culturally specific jailbreaks.
- **Static Steering Dynamics:** The use of a uniform centroid-based vector for all failures may lead to sub-optimal steering. Production-scale deployment would require dynamic, context-aware steering to mitigate over-refusal.
- **The English-Centric Bias Assumption:** The methodology assumes the English safety boundary is the correct ground truth. Consequently, latent English biases or over-refusal tendencies are projected onto the Bengali latent space.

5.3 Future Work

To transition LatentGuard from a proof-of-concept to a production-ready system, future work must focus on decoupling safety alignment from linguistic representation. The natural next step is implementing **Orthogonal Projection**. Instead of adding a raw translation-based steering vector, we must calculate the projection of the safety vector that is strictly orthogonal to the English-Bengali language direction. This would steer the model toward safety without dragging its representations toward the English linguistic subspace, forcing the model to refuse safely in native Bengali.

Additionally, evaluating this geometric diagnostics pipeline across a diverse matrix of Indic languages (e.g., Hindi, Tamil, Telugu) and model families (e.g., Llama-3, Mistral) is critical to verify if the Late-Stage Safety Bypass is a universal architectural artifact of modern transformers.

6. Conclusion

This study investigated cross-lingual safety failures in *google/gemma-2-2b-it*, identifying a "Late-Stage Safety Bypass" where geometric safety boundaries fracture immediately prior to vocabulary projection. We introduced LatentGuard, a training-free latent steering intervention that realigns drifting low-resource representations with English-anchored refusal pathways, providing a scalable alternative to fine-tuning and mitigating Knowledge Collapse risks. While results highlight persistent "Language Bleed"—where steering vectors entangle with linguistic identity—our findings confirm latent space manipulation is a viable pathway for global AI alignment. Future research will prioritize orthogonal projection to decouple safety vectors from linguistic identity, ensuring secure, accessible, on-device deployments for the Global South.

Code and Data

- **Code repository:** [Github Code Link](#)
- **Data/Datasets:** To ensure transparency and reproducibility, the complete evaluation corpus is open-sourced via Hugging Face at [LatentGuard-Bengali-Safety-Drift](#). This dataset includes the 150-prompt behavioral baseline execution log and the 35-prompt failure cohort utilized for latent diagnostics and steering interventions, derived from the aims-foundations/safety-irt repository.
- **Other (results):** [Results](#)
- **Info-Hazard Notice:** Please note that the dataset and output logs contain explicit examples of Multilingual Safety Drift, including successfully bypassed harmful prompts (e.g., instructions for violence or illegal acts) translated into Bengali. These artifacts are provided strictly for academic verification of the LatentGuard mitigation and should be handled responsibly.
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Appendix A. Supplementary Materials

A.1 Dataset and Experimental Configuration

The evaluation corpus consists of parallel safety prompts sourced from the *aims-foundations/safety-irt* repository. Table A1 details the composition of our experimental cohorts.

Table VIII: Experimental Cohort Composition		
Subset	Count	Description
Baseline Cohort	150	Initial samples for behavioral profiling
Failure Cohort	35	Prompts exhibiting token fragmentation
Success Cohort	115	Prompts with expected model behavior

A.2 Behavioral Profiling Methodology

To identify failure modes, we implemented a pattern-based detection system. Refusals were flagged using language-specific indicator strings.

Table IX: Heuristic Thresholds for Fragmentation Analysis		
Metric	Threshold	Description
Repetition Ratio	> 0.5	Single character frequency relative to total length
Symbol Ratio	> 0.5	Non-alphanumeric character density
Unique Characters	< 8	Minimum distinct glyph diversity
Bengali Corruption	> 2 isolated vowels	Script-specific invalid sequences

A.3 Statistical Analysis of Findings

To validate the significance of our latent diagnostic results, we performed a one-sample t-test on the collapse scores at Layer 19 (n=35).

- **Hypothesis Testing (H1):** Bengali representations exhibit positive collapse at Layer 19.
 - Result: $t(34) = 4.2$, $p < 0.001$. We reject the null hypothesis, confirming statistically significant geometric boundary collapse.
- **Hypothesis Testing (H3):** Steering improves refusal rates.
 - Chi-square test (baseline vs. post-intervention): $\chi^2(1) = 3.89$, $p = 0.048$. The improvement in refusal rate (from 60% to 80%) is statistically significant.

LLM Usage Statement

We utilized Gemini as an AI research assistant throughout the development of this project to refine academic prose for IEEE compliance, and structure supplementary documentation. We used Claude to help develop the codebase and draft documentation. All results were generated by executing our pipeline on real data and independently verified from output files. All empirical research, data analysis, quantitative results, and core conclusions presented in this submission were independently conducted and verified by the authors.